ANY ZARISKI COVERING IS A PH COVERING

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [2639, 2645] and proof(s) [[2647, 2651]]; this spec covers the complete environment instead (statement 2639-2645, proof span(s) [(2647, 2651)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Editorial note. Definitional given the accepted refinement lemma.

Source reference. Original reference 0DBT: sources/stacks-chow/tags.txt, source lines 16708–16708. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0DBF: sources/stacks-chow/tags.txt, source lines 16695–16695. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *A Zariski covering is a ph covering*[1].

Proof. This is true because a Zariski covering of an affine scheme can be refined by a standard ph covering by Lemma 0DBF. □

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[1]We will see in More on Morphisms, Lemma 0DBT that fppf coverings (and hence syntomic, smooth, or étale coverings) are ph coverings as well.